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Kinetics of thermophilic batch anaerobic digestion of thermal hydrolysed waste activated sludge

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ABSTRACT

Enhancing waste activated sludge (WAS) anaerobic digestion by thermal pretreatment has become of high interest. However, thermal treatment has been mainly combined to mesophilic anaerobic digestion. This paper presents the combination of sludge thermal pretreatment (110, 165 and 220 °C) and batch thermophilic anaerobic digestion (55 °C). Optimal conditions of thermal pretreatment were shown to be 165 °C, involving a chemical oxygen demand (COD) and volatile solids (VS) solubilisation of 18 and 15% and a biodegradability increase from 47 to 61%. Treatments at 165 °C were carried out in electric and steam modes and no significant difference on the impact of heating mode on sludge anaerobic biodegradability was observed. Moreover, it may be recommended not to carry out successive batch experiments to assess thermophilic BMP of sludge as accumulation of volatile fatty acids (VFA), particularly propionate, and a decrease of VFA uptake rates may occur. However, thermal pretreatment at 165 °C allowed the decrease of propionate accumulation and an higher methane production.

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1. Introduction

Biological processes are developed as the main process to improve efficiently the quality of the effluent in municipal wastewater treatment plants. Waste activated sludge (WAS), as a by-product, are generated in large and increasing quantities. Since the development of the biogas sector in France, following the July 2006 publication of new sufficiently attractive prices, WAS is more and more considered as renewable energy source and becomes an interesting substrate to anaerobic digestion. However, WAS anaerobic digestion is more difficult than for primary sludge and, with usual technologies nowadays available, only approximately 20–30% of the sludge organic matter are mineralized [1].

Anaerobic digestion process follows four major steps: hydrolysis, acidogenesis, acetogenesis and methanogenesis. Hydrolysis is a slow step and is considered as the rate-limiting step of the overall process in the case of sludge degradation [2]. During hydrolysis step, organic compounds, such as polysaccharides, proteins and fats, are hydrolysed by extracellular enzymes to monomer or dimer compounds.

Over the last years, pretreatment steps, such as physical treatment with bead mill, sonication or high-pressure homogenizer, biological treatment with enzymatic hydrolysis, chemical treatment with alkaline addition were applied to improve hydrolysis of particulate organic matter and substantially biodegradability of sludge. Thermal treatment has been widely combined to anaerobic digestion performed in the mesophilic range and this resulted in an increase of biogas production and of the kinetic rates [3] and energy costs can be covered by biogas production [4,5]. Climent et al., Bougrier et al. and Jeong et al. [6–8] underlined the impact of solubilisation of particulate organic matter on the biogas production enhancement during anaerobic digestion. Another way to increase anaerobic digestion performances consists in operating digestors in thermophilic mode. Indeed, thermophilic anaerobic digestion allows to improve degradation rates, hence shorter retention times and provides an enhancement of the biogas production [9]. However, very few studies deal with the combination of thermal pretreatment and thermophilic anaerobic digestion of sludge [10–12].

The objective of this work was to carefully analyse the impact of thermal pretreatment on thermophilic anaerobic digestion of WAS. Efficiency of thermal pretreatment was evaluated by solubilisation of COD and volatile solids (VS). Methane production and kinetics of volatile fatty acids (VFAs) and soluble chemical oxygen demand (COD) were measured during batch anaerobic biodegradation in order to investigate in details the hydrolysis of pretreated and untreated organic matter. Moreover, electric and steam modes

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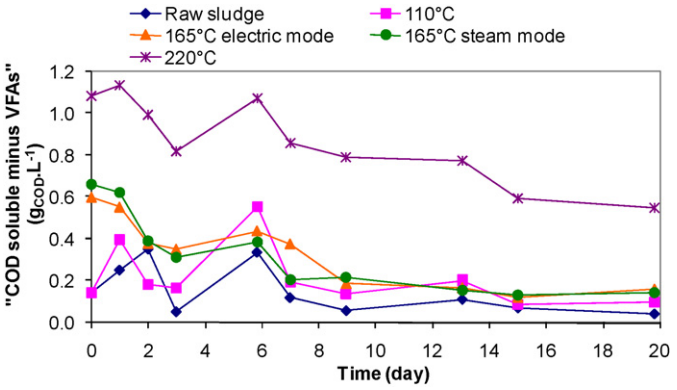

Fig. 2. Soluble compound concentrations during batch anaerobic digestion of each sludge (batch 4).

carried out in a same reactor for each substrate in order to evaluate the inoculum adaptation. The experiments were monitored during 22 days.

Fig. 2 represents the measured $COD_{soluble}$ minus VFAs in batch 4. It can represent the transfer of compounds from the particulate phase to the soluble phase, i.e., the behaviour of hydrolysis, thus the production of soluble compounds (<0.45 μm), before their degradation by acetogenesis and acidogenesis.

Fig. 3 represents methane production and VFA concentrations for batch 2, 3 and 4 for raw sludge and sludge treated at 110, 165 °C and 220 °C. VFA concentrations were not monitored for batch 1. It is important to define the VFA concentration variations monitored during experiments; they result from the VFA production by acidogenesis and from the VFA uptake by acetogenesis and aceticlastic methanogenesis. For example, an accumulation of acetate can be due to the step of acetate production faster than the step

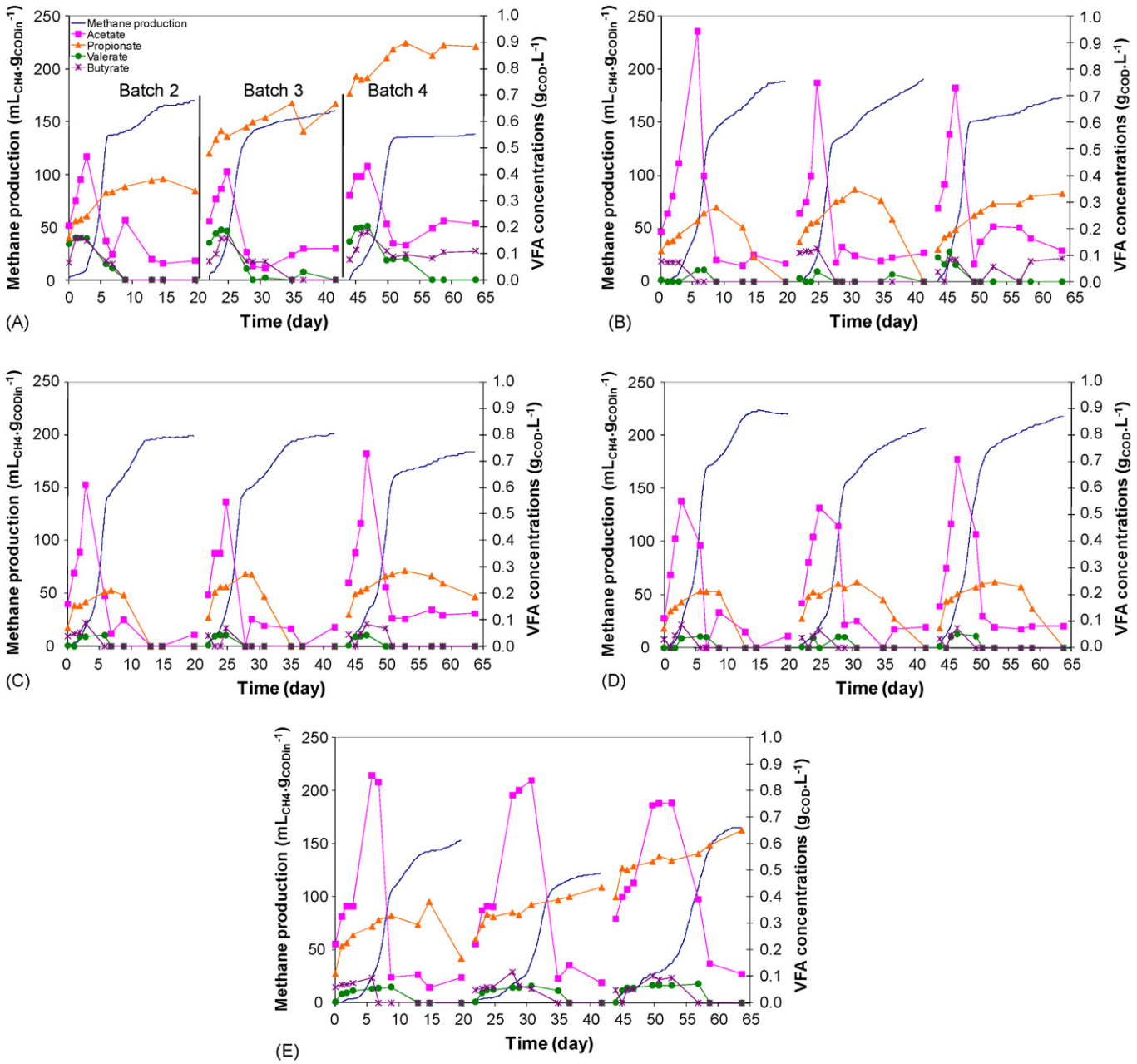

Fig. 3. Methane production and VFA concentrations during batch anaerobic digestions of (A) untreated sludge, (B) pretreated at 110 °C, (C) pretreated at 165 °C in electric mode, (D) pretreated at 165 °C in steam mode and (E) pretreated at 220 °C.

of acetate uptake. These values can bring interesting information into the understanding of mechanisms involved in the methane production.

The soluble COD concentration variations (Fig. 2) of sludge pretreated in optimal conditions (165 °C) showed a direct degradation of soluble organic matter and no significant accumulation of soluble COD. Thus, pretreatment allowed a better accessibility of organic matter, which was directly available for the biological steps, and could minimise the limiting effect of biological hydrolysis.

For the sludge pretreated at 220 °C, despite a large solubilisation of COD (27%), soluble compound concentration at the end of experiment was high and decreased very slowly. It seems to indicate that compounds, formed at 220 °C (e.g. memanoidins and Amadori compounds) were hardly biodegradable.

In the case of raw sludge and pretreated at 110 °C, a first phase of degradation of particulate organic matter from day 0 to day 3 and a second phase of degradation from day 3 to day 6 were observed. The significant accumulation of soluble components seems to lead to the conclusion that two types of particulate matter constitute the sewage sludge: one fraction is quickly hydrolysed whereas the second fraction is more slowly hydrolysed. These two fractions are less significant on the soluble COD concentration variations of sludge pretreated at 165 °C. But they can be clearly observed for all sludge samples in Fig. 2.

Indeed methane production curves showed two phases (Fig. 3). A first one, with a fast production corresponding to the degradation of readily accessible organic matter, like monomers and dimers compounds or exopolymers, lasted from day 2 to day 6. It is also characterised by the production and the accumulation of VFAs (from day 0 to day 4). The second phase, with a slower production, corresponding to the degradation of hardly accessible organic matter, like particulate macromolecules strongly linked in sludge structure or compounds located inside the cells lasted from day 6 to days 12–14. Finally, methane production was null or very low from day 14 to day 22.

The variations of acetate and propionate concentrations could be associated to the first and the second phases of methane production, respectively. Batch experiment 2 with sludge treated at 165 °C in electric mode (Fig. 3C) could confirm this hypothesis. It was observed that the first phase of methane production was mainly associated to the acetate uptake, from the day 0 to day 6. During this phase, the highest methane volume was produced and the acetate concentration decreased strongly. The second phase of methane production was associated to the propionate uptake. Indeed the start-up of the second methane production phase with the uptake of propionate at day 7 could be observed and a plateau was reached at day 12 when the VFA uptake was complete. According to Vavilin et al. [22], the acetogenesis and methanogenesis can be the rate-limiting steps in anaerobic digestion of a complex substrate at a high organic loading. These observations seem to show that the acetate and propionate degradation steps can also be considered as the limiting step for the anaerobic digestion of a complex substrate (waste activated sludge) with an organic loading of 0.5 g_{COD} per g_{VS} of inoculum.

As discussed above, total methane production increased with thermal pretreatment temperature. However, a significant decrease of methane production for raw sludge was observed through successive batch experiments (Fig. 3A). This seems to be linked to the propionate accumulation. Thus successive thermophilic digestion batches of raw sludge may lead to an accumulation of propionate as already observed by Speece et al. [23].

In the case of sludge treated at 220 °C, the first phase of methane production, which was from day 0 to day 6 for the batch 2, had a tendency to increase in time through consecutive batch experiments, while a small methane production and a slow degradation rate of acetate were observed. This seems to indicate that the

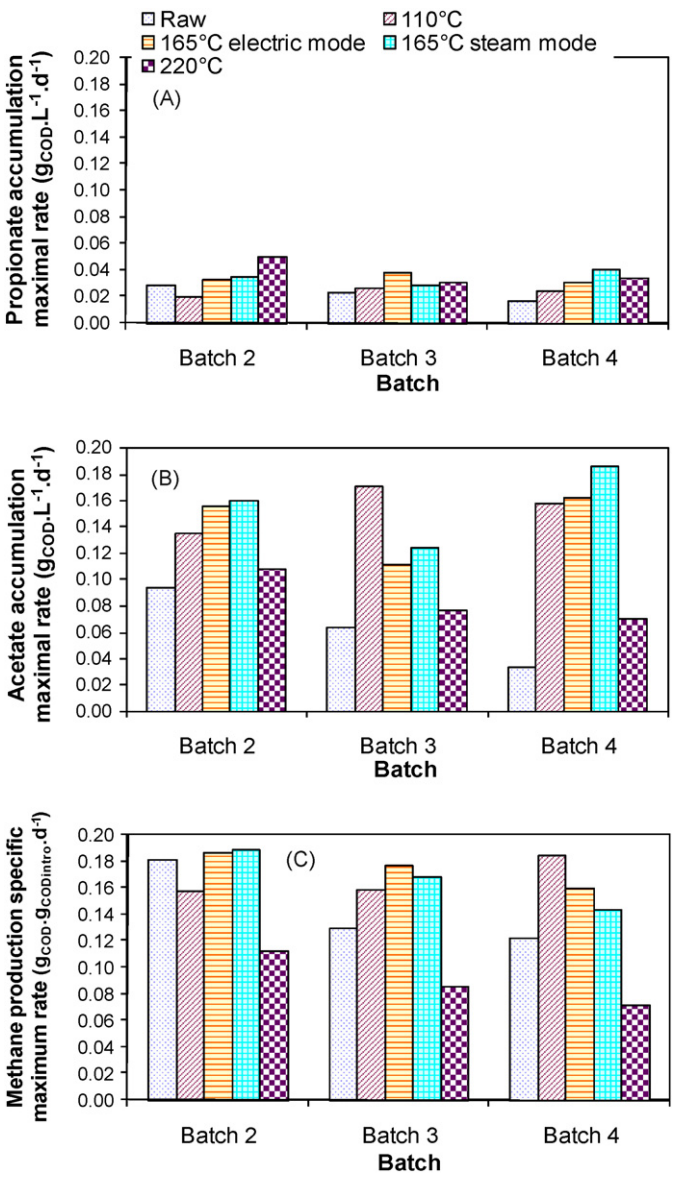

Fig. 4. Acetate (A), propionate (B) accumulation maximum rates and methane (C) production specific maximum rates with different conditions of pretreatment through batch experiments.

hardly biodegradable components produced at 220 °C are slowly biodegradable. Thus the results can confirm that the melanoidins and Amadori compounds are hardly and slowly biodegradable. The enhancement of methane production observed during batch 4 could arise from an overestimation of produced volume (resulting from very slowly biodegradable compounds from previous experiments) and not from an adaptation of inoculum.

The values of propionate and acetate accumulation maximum rates, K expressed in $g_{COD}\ L^{-1}\ day^{-1}$, can be used to accurately evaluate the impact of thermal pretreatment on the mechanisms of acetogenesis and methanogenesis steps (Fig. 4A and B). This parameter was determined from the slope of the linear regression line associated to the accumulation phase of propionate and acetate from day 0 to the day corresponding to the maximal accumulation of each batch experiment.

The values of $K_{propionate}$ did not show statistically significant differences (Fig. 4A). However, important differences were observed between the variations of propionate concentrations of each sludge with an accumulation of propionate in the case of the raw sludge

and the sludge pretreated at 220 °C (Fig. 3A and E), which could be correlated to the values of $K_{acetate}$. Indeed, these propionate accumulations were associated to a low acetate accumulation maximum rate which may be associated to a low acetate production rate (Fig. 4B). The values of $K_{acetate}$ were below 0.10 g_{COD} L^{-1} day^{-1} when the propionate accumulation took place. Thus, propionate accumulations involved a partial inhibition of acidogenesis.

Fig. 4C presents the maximum methane specific production of each sludge through consecutive batch experiments, K_{CH_4} being expressed in g_{COD} $g_{CODintro}$ day^{-1}. This was determined from the maximum methane production specific rate throughout experiment time.

If we consider raw sludge, K_{CH_4} decreased which was followed by the decrease of final methane volume through consecutive batch experiments (Fig. 3A). This was associated to a strong propionate accumulation and a decrease of $K_{acetate}$. This comment could confirm that a high propionate concentration can involve a partial inhibition of acidogenesis and methanogenesis steps. This was already underlined by Hyun et al. [24] who observed an inhibition of acetate degradation by a high propionate concentration (1 g_{COD} L^{-1}) during the anaerobic digestion of propionate-enriched mixed cultures at 35 °C.

The values of $K_{acetate}$ through consecutive batches for the sludge treated at 220 °C were low with a value lower than 0.10 g_{COD} L^{-1} day^{-1}, and a large production of acetate was observed with a slow degradation of acetate, which was associated to a low K_{CH_4}. Thus the produced compounds (melanoidins and Amadori compounds) had a negative effect on acetate degradation and methane production. These results confirmed that a fraction of solubilised organic matter were hardly and slowly biodegradable.

For the sludge samples pretreated at 110 °C and 165 °C (both modes), $K_{acetate}$ values were not significantly different and were stable. However, they showed important and statistically significant higher values than for raw sludge. For example in batch 2, $K_{acetate}$ was equal to 0.0939 g_{COD} L^{-1} day^{-1} and 0.1593 g_{COD} L^{-1} day^{-1} in the case of digestion of raw sludge and sludge pretreated at 165 °C in steam mode, respectively. This enhancement was observed in each batch experiment. Moreover, the propionate accumulations were less important than in the case of raw sludge and sludge pretreated at 220 °C. Thus, the solubilisation induced by the thermal pretreatment has an important impact on the step of acetate production from propionate and soluble components. Fig. 2 showed that the thermal pretreatment allowed a decrease of the limiting effect of the anaerobic digestion hydrolysis step by solubilising a large part of organic matter, which is more available by the anaerobic microorganisms, in particular in the case of sludge pretreated at 165 °C in both modes. The solubilised organic matter was directly degraded, allowing higher acetate and methane production rates, no or low propionate accumulation and a higher quantity of degraded organic matter.

The thermal hydrolysis enhanced the availability of organic matter. Thus, the solubilisation allowed to bring an available substrate in high quantity and it could allow maximising acidogenesis by a better development of acidogens and no propionate accumulations.

A second hypothesis is possible to explain the non-accumulation of propionate in the case of sludge pretreated in optimal conditions: anaerobic digestion can be stimulated by element trace metals. Indeed, Zitomer et al., Kim et al. and Speece et al. [25,26,23] showed that the supplementation of trace metals, like cobalt, iron and nickel had a positive effect on the degradation of propionate and acetate. Zitomer et al. [25] obtained a statistically significant increase of degradation of propionate and acetate for 77% of tested thermophilic biomass samples, upon supplementation with either nickel, cobalt or iron. The propionate degradation rate more frequently increased upon nutrient addition, especially in thermophilic systems, whereas the acetate degradation rate increased less frequently. In the sludge, the total metal concentrations may appear adequate but the metals may not be present in bioavailable form for the microorganisms. Thus, the disintegration of the sludge structure, induced by the thermal hydrolysis, could allow to make the trace metals more available for the microorganisms and could enhance the degradation of propionate. However, this hypothesis could not be validated since the measurements of trace metals were not realised.

4. Conclusions

Optimal conditions of thermal pretreatment were obtained at 165 °C, involving a COD and VS solubilisation of 18 and 15% and a biodegradability increase from 47 to 61%. At 220 °C, although a large solubilisation of particulate organic matter occurred, sludge biodegradability was lowered than other sludges (42%). Thus, a too high temperature treatment leads to the production of slowly and hardly biodegradable compounds, like Amadori compounds and melanoidins.

The Newman–Keuls test showed that the difference of BMP between treatments at 165 °C (in electric and steam modes) was not significant. Thus, the laboratory thermal hydrolysis carried out in electric mode represents properly industrial thermal hydrolysis carried out with steam injection.

For all untreated and pretreated sludge, successive thermophilic batch digestion tests were not necessary to evaluate maximal methane production, as an adapted inoculum was used and VFA accumulations occurred through successive batches. One batch test, monitored until VFA concentrations reaches zero, seems sufficient to measure the maximal methane production of WAS.

Methane production kinetics were strongly linked to the acetate and propionate kinetics. The results showed that the degradation of propionate and acetate had a limiting effect on the methane production. The acetate degradation was associated to the first methane production phase and the propionate degradation was associated to the second methane production phase.

The thermal hydrolysis could minimise the limiting effect of hydrolysis due to a large solubilisation of organic matter, thus improved the availability of particulate organic matter and improved the acetate production rate, inducing a better conversion of propionate to acetate that avoided propionate accumulation.

Further work will concern the modeling of these results by ADM1, a complete model for anaerobic digestion. Special attention will be paid to the description of the hydrolysis which is the limiting step of anaerobic digestion of solid organic matter.

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